

Analysis group meeting

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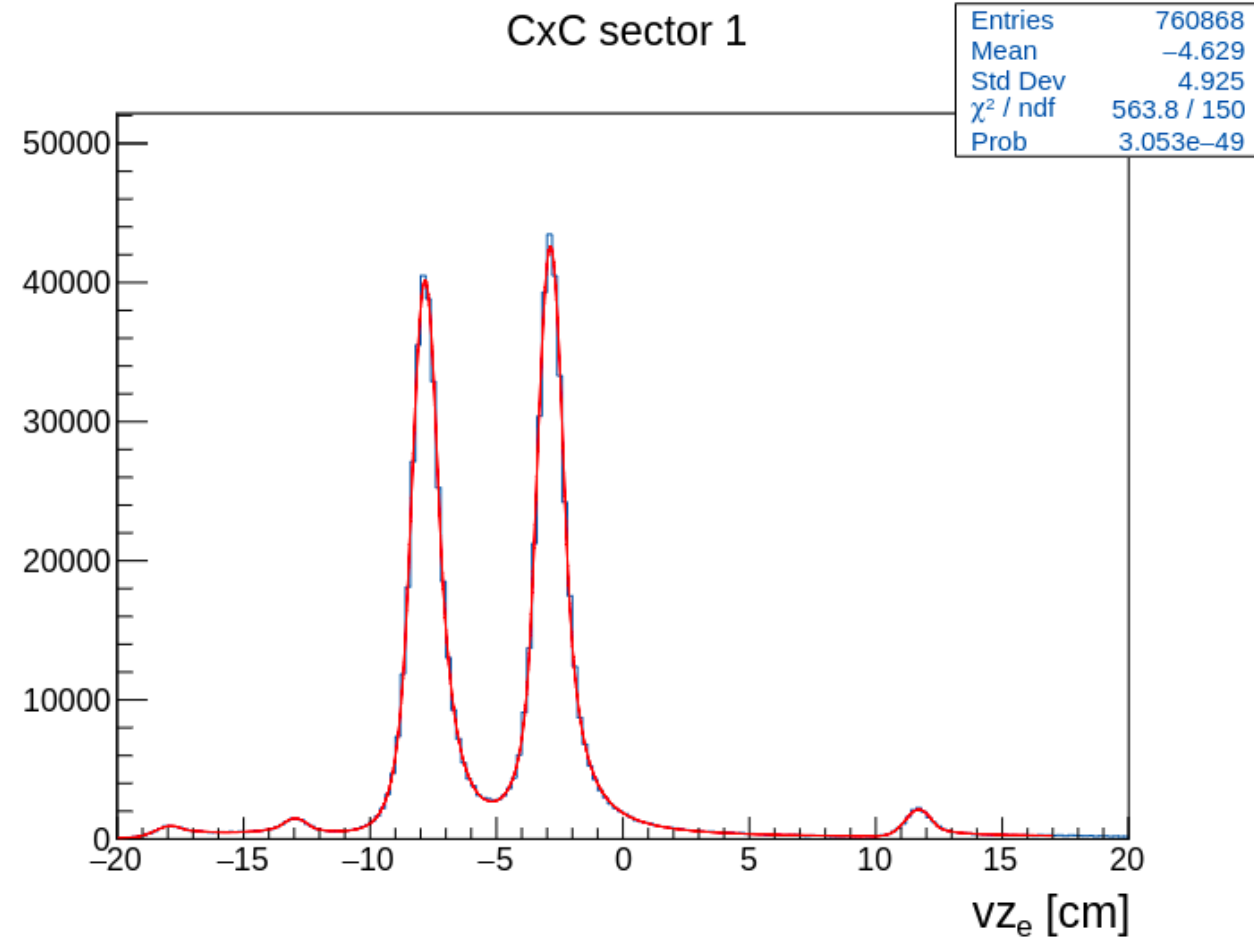
Cooking chef

- Produce timelines with 418 runs yesterday.
- Some issues with the disk quota :
 - The farm_out is full, due to the ShortEmpRand workflow
 - The run 18316 generate .out files with a size around 2.2 G
 - Error : no more : too many combinations..... 70 M lines
- Workflow ShortEmpRand: 44.6% (percentage of runs with hist folder fill)
- Workflow LD2: 99.3%
- Workflow CxC: 64.0%
- Workflow ShortEmpRand: 64.7%

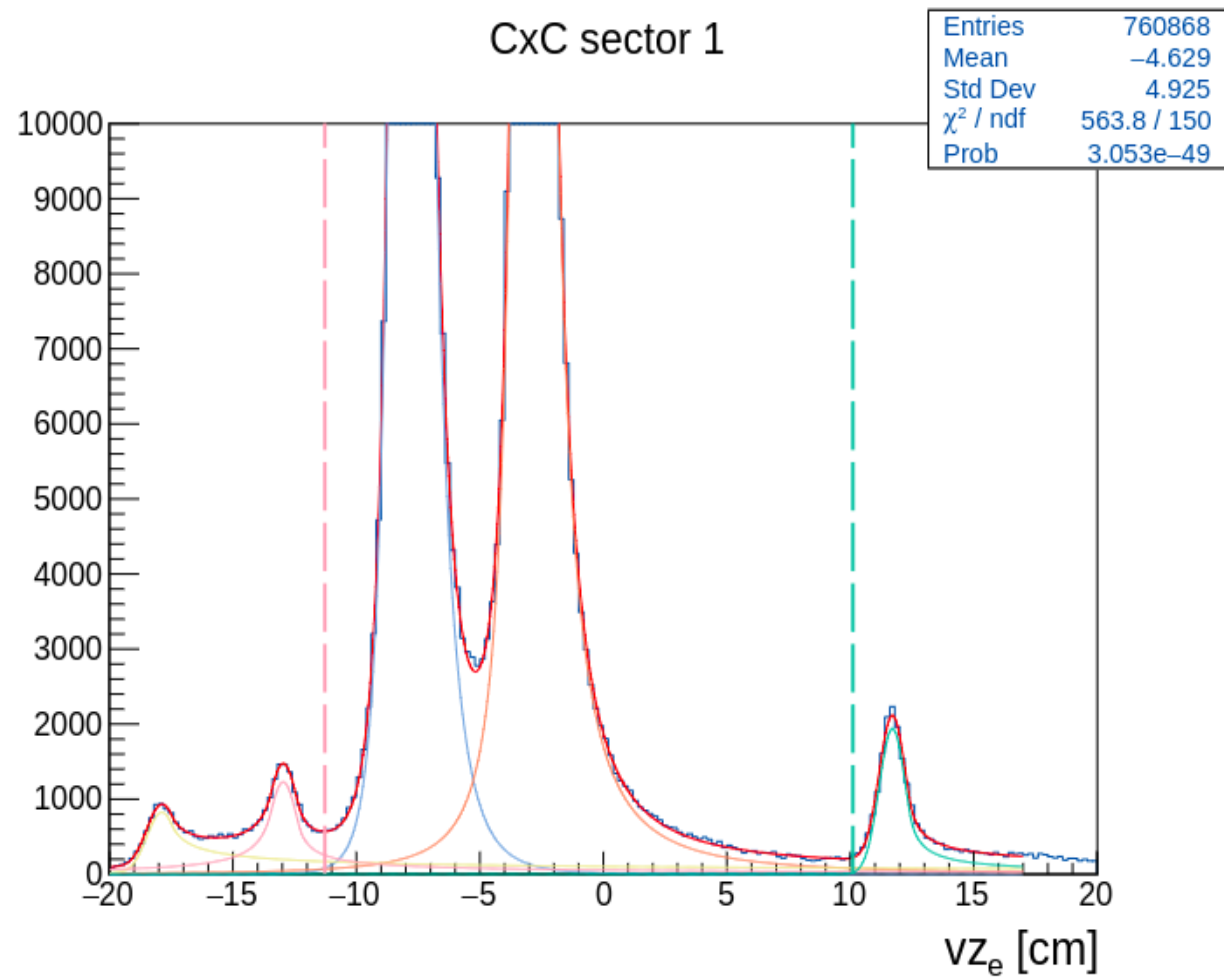
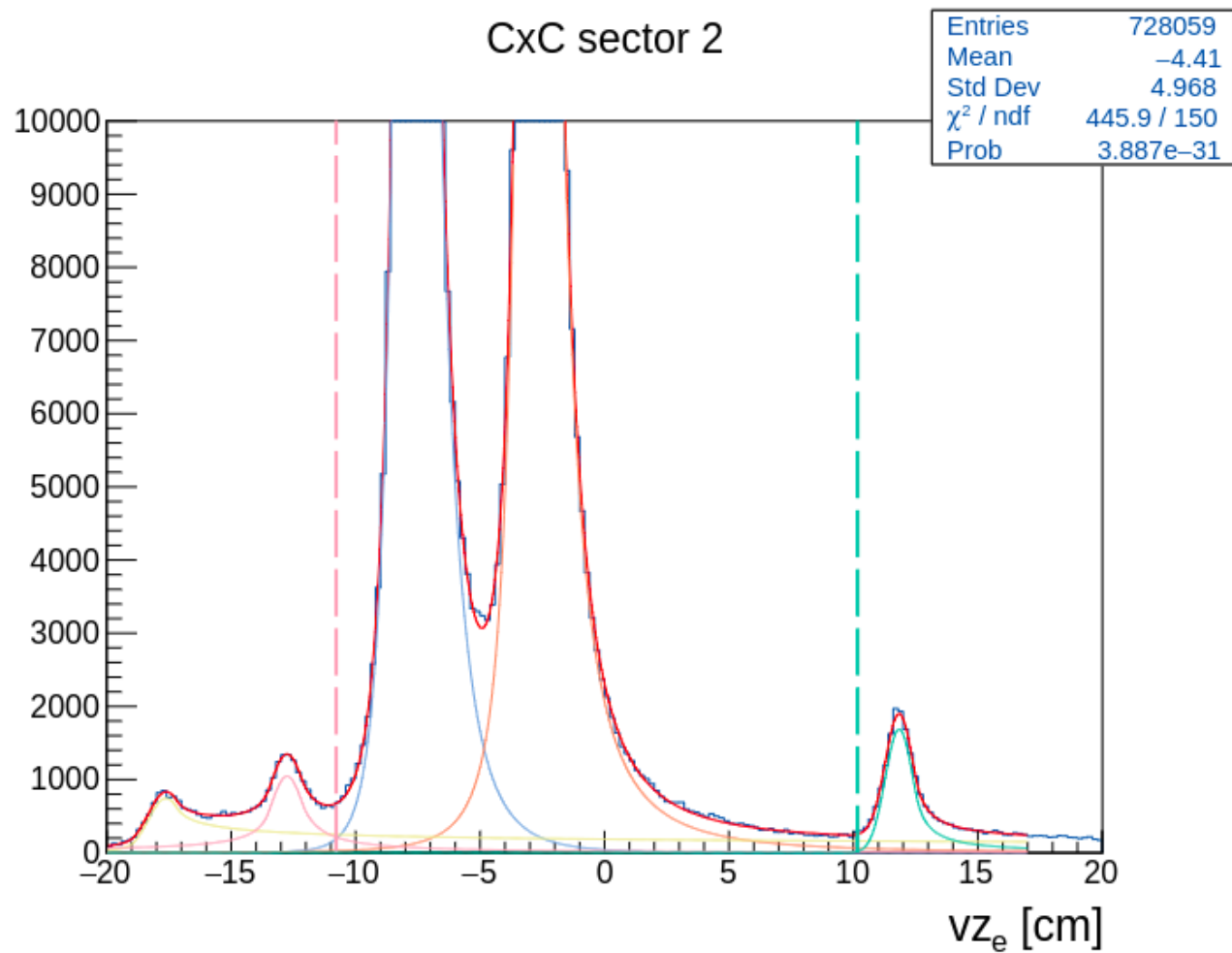
RG-D analysis: vz study

- First use run 018344 (target CxC)
- Try with gaussians, but fail
- Use 5 double-sided Crystal Ball (DSCB) function
- Very instable, a small modification in the initial parameters, fit fail.

$$N \cdot \begin{cases} e^{-0.5t^2} & \text{if } -\alpha_{\text{low}} \leq t \leq \alpha_{\text{high}} \\ e^{-0.5\alpha_{\text{low}}^2} \left[\frac{\alpha_{\text{low}}}{n_{\text{low}}} \left(\frac{n_{\text{low}}}{\alpha_{\text{low}}} - \alpha_{\text{low}} - t \right) \right]^{-n_{\text{low}}} & \text{if } t < -\alpha_{\text{low}} \\ e^{-0.5\alpha_{\text{high}}^2} \left[\frac{\alpha_{\text{high}}}{n_{\text{high}}} \left(\frac{n_{\text{high}}}{\alpha_{\text{high}}} - \alpha_{\text{high}} + t \right) \right]^{-n_{\text{high}}} & \text{if } t > \alpha_{\text{high}}, \end{cases}$$



RG-D analysis: vz study

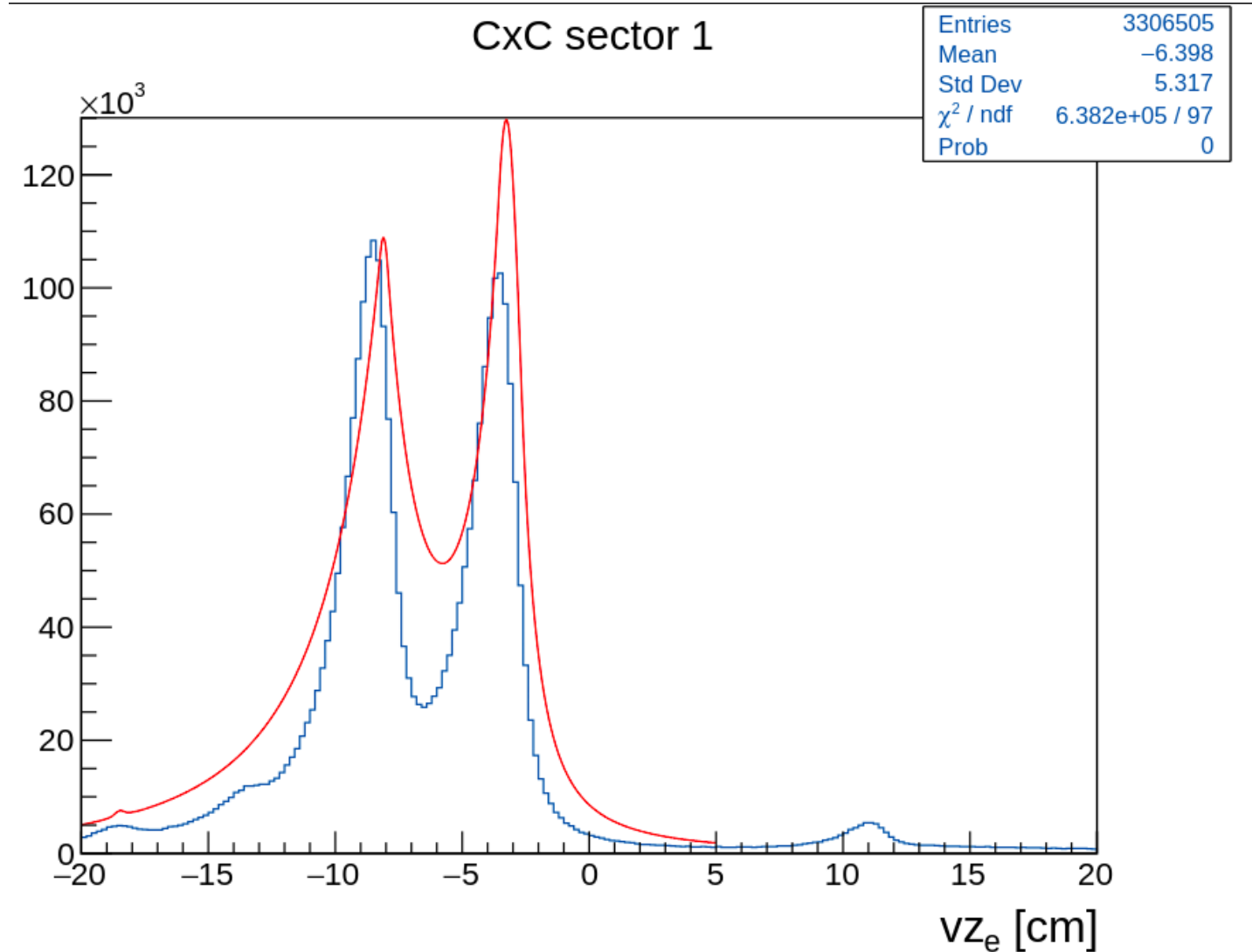


RG-D analysis: vz study

Colonne 1	T _r Sector 1	T _r Sector 2	T _r Sector 3	T _r Sector 4	T _r Sector 5	T _r Sector 6
mean_exit_LD2	-12.9604	-12.7539	-12.4224	-12.2442	-12.1534	-12.6273
sigma_exit_LD2	0.561525	0.657872	0.61651	0.696992	0.674343	0.6309
cut position_exit_LD2	-11.27	-10.7803	-10.5729	-10.1532	-10.1304	-10.7346
mean_entrance_scat	11.7051	11.8383	12.168	12.3334	12.481	12.0623
sigma_entrance_scat	0.533547	0.556068	0.524564	0.568106	0.568106	0.511755
cut position_entrance_scat	10.1045	10.1701	10.5943	10.6285	10.7767	10.5271

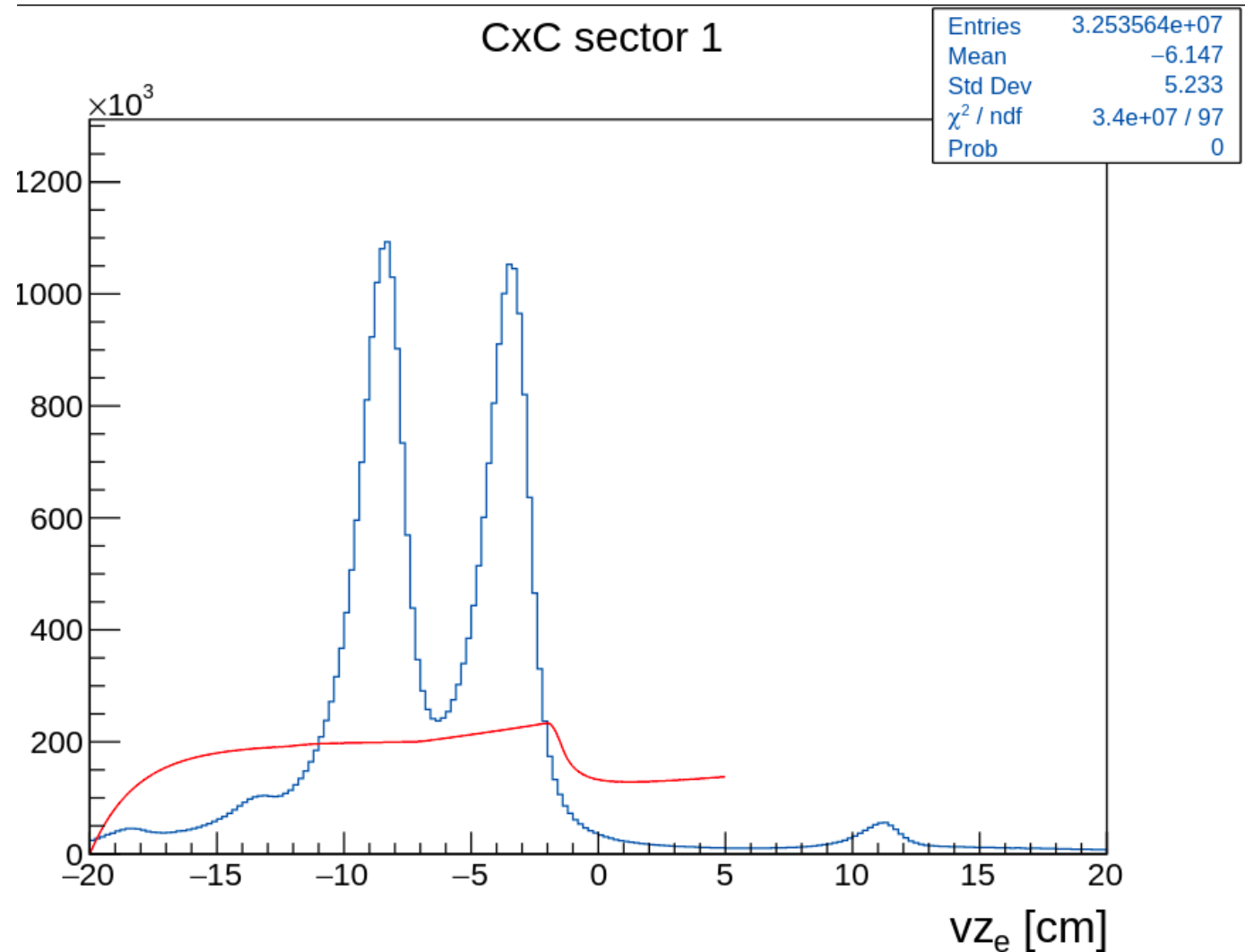
RG-D analysis: vz study

- But problem with other runs, different shape
- Run 018498, target CxC
- Haven't been able to fit with the same function.



RG-D analysis: vz study

- Shape for the all stats (1077 files, target CxC)



RG-D analysis: Matt analysis crosscheck

- All CxC runs : 018339, 018340, 018341, 018342, 018343, 018344, 018346, 018440, 018441, 018442, 018443, 018444, 018445, 018475, 018498, 018524, 018756, 018850
- path: /cache/hallb/scratch/rg-d/production/Bspot/v5dstCxC/dst/recon/
- Good electron (electron than pass the status, pid, chi2 and vz cut and at least one pion+ pass their cuts and at least one pion-):
14320539 | 14.32054 M (me, Matt)
- Good pion+ (pion+ than pass the pid, chi2 and a good electron and at least one pion-): 17021464 | 17.02146 M
- Good pion- (pion- than pass the pid, chi2 and a good electron and good pion+): 17056311 | 17.05631 M

RG-D analysis: Matt analysis crosscheck

- Number of rho0 that pass the W, zh, t, Q2 : 107556 | 107556
- For the Q2 bins with W, Lc, t & Zh cuts
 - Q2 1-2 = 3578 | 3578
 - Q2 2-2.5 = 2561 | 2561
 - Q2 2.5-3 = 1748 | 1748
 - Q2 3-3.5 = 988 | 988
 - Q2 3.5-4.5 = 885 | 855
 - Q2 4.5-6 = 370 | 370
- Difference in the number of rho0 for each Q2 bin.
 - Need to check the integral.

ALERT: AI for track finding

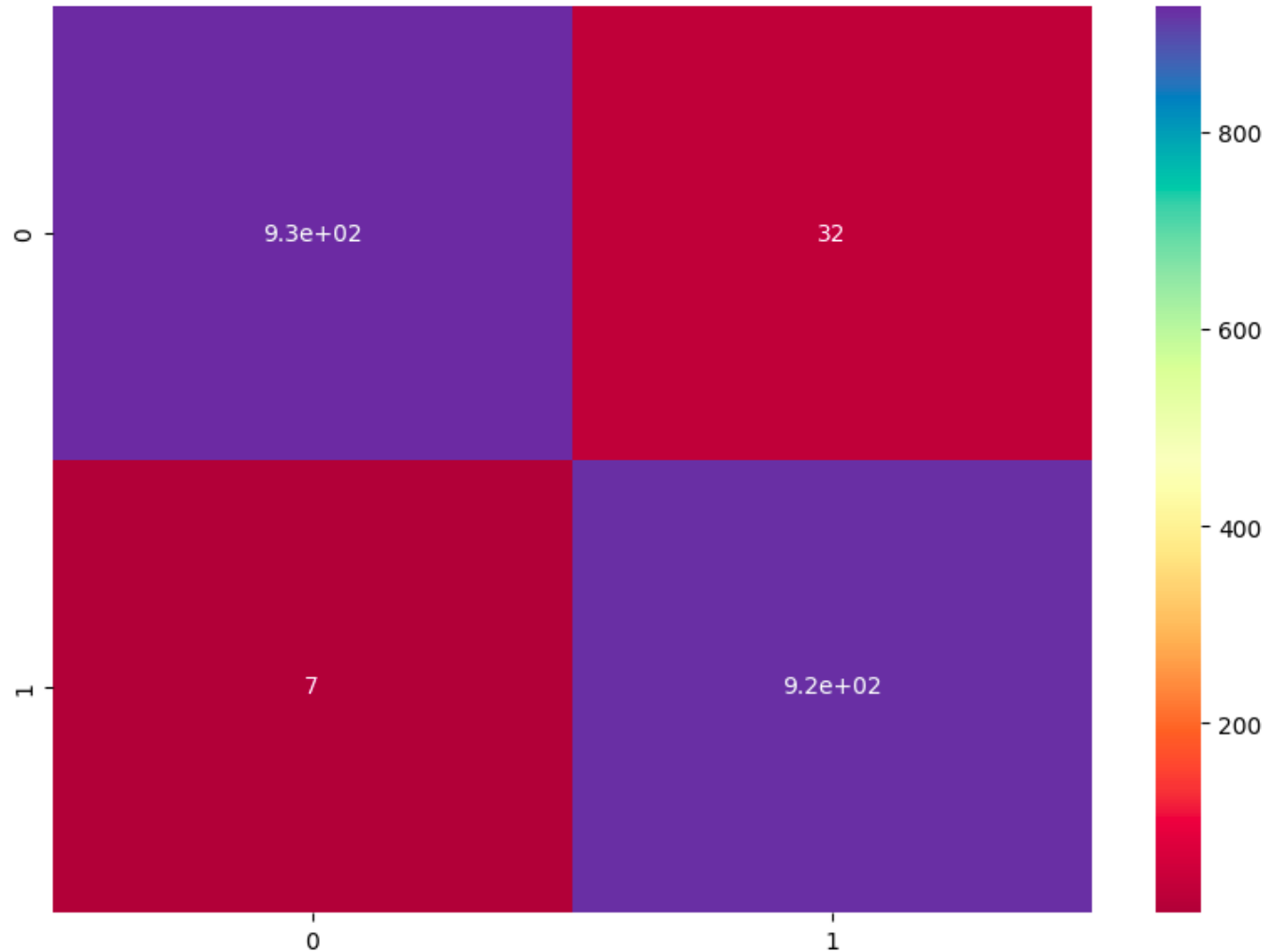
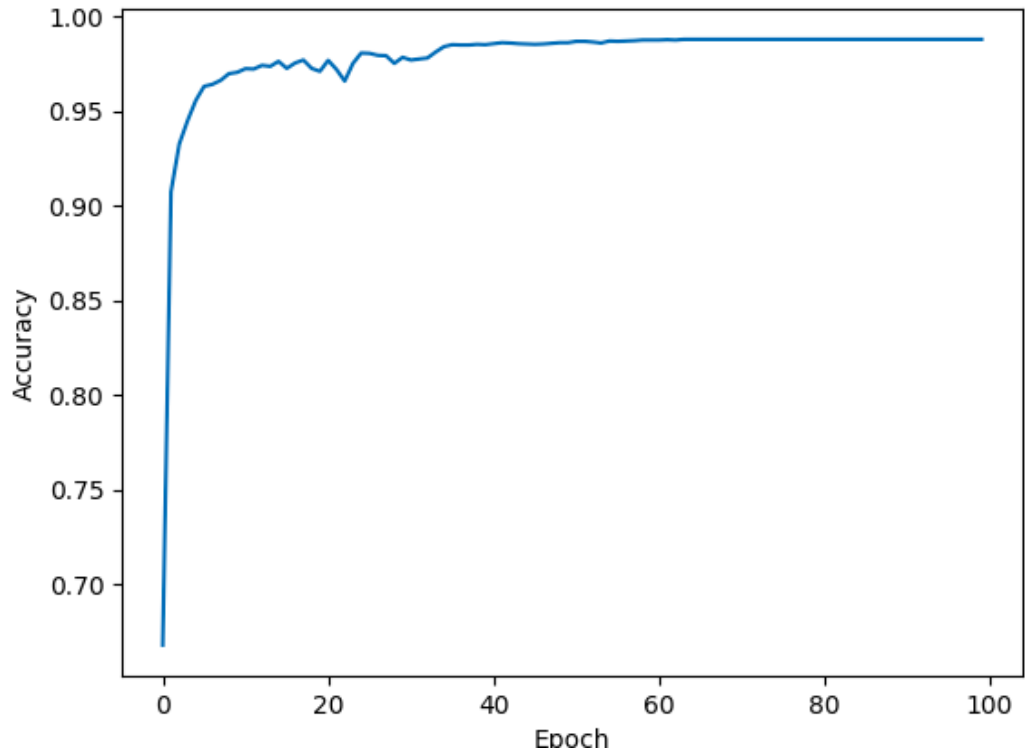
- Start working on the track finding for ALERT.
- Generate 50-50 sample files (good/bad tracks)
 - Simulate 10K events in gemc
 - bad tracks were constructed by randomly interchanging upto 3 preclusters from the previous event
- Use pytorch lightning for the model and training
 - Model : 16 inputs -> 32, 32, 32, 1 output, use x,y for the PreCluster
- Get a accuracy/score of 98%, close to what Udi get.

ALERT: AI for track finding

Confusion matrix:

0 bad track

1 good track



ALERT: AI for track finding

- To compute the efficiency :
 - After the model is train using the 50-50 sample
 - Generate some randoms hits (up to 5)
 - Generate all possible combination with the PreCluster (only one PreCluster per layer)
 - Check how often the model find the good track (and only the good track) in a event = efficiency.
- Get a efficiency of 76%.

To do list

- Cooking chef:
 - Continue to make and deploy the timelines.
 - Looking at the calibration workflow.
- RG-D analysis:
 - For vz analysis: find a way to fit the distribution for all the runs.
 - For Matt analysis crosscheck:
 - Check the integral to get the rho0 number.
 - Cross check the numbers for LD2.
 - Made skim per run
 - Plot for the electron vz vs theta, phi (lambda analysis)
 - Understand the tail of z_h
- ALERT AI:
 - Use a larger sample to train.
 - Test the model if I use the hits
 - MLP accuracy if the random hit are 1/2/3/4/... wire away of the track
- Gabriel (ALERT engineer) ask me if I can help him with cabling ALERT. It is 2 days between 8-17 october (he is no sure when exactly.)